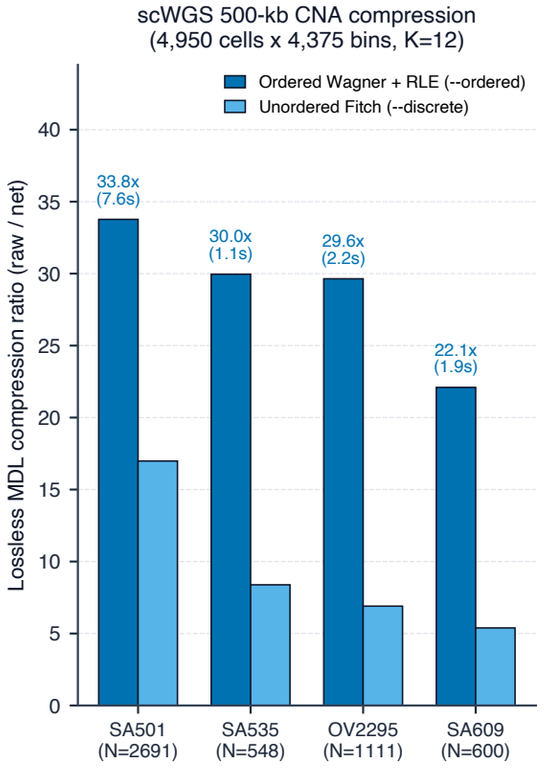
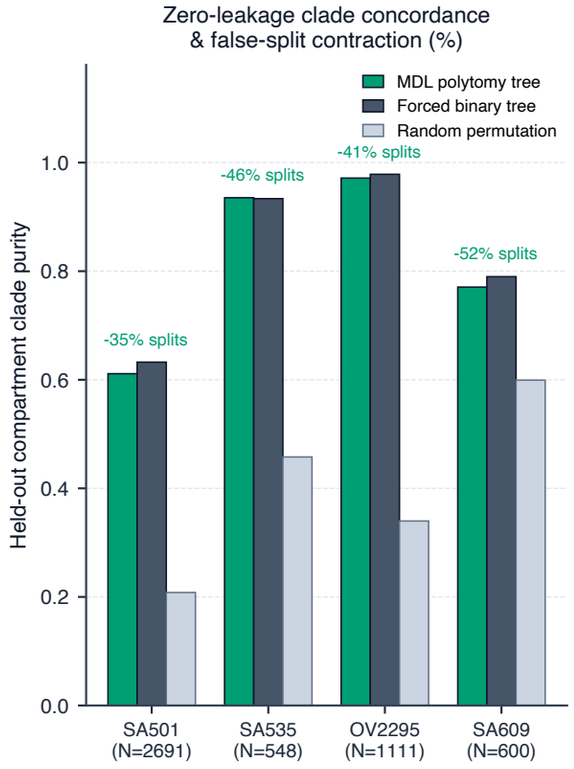
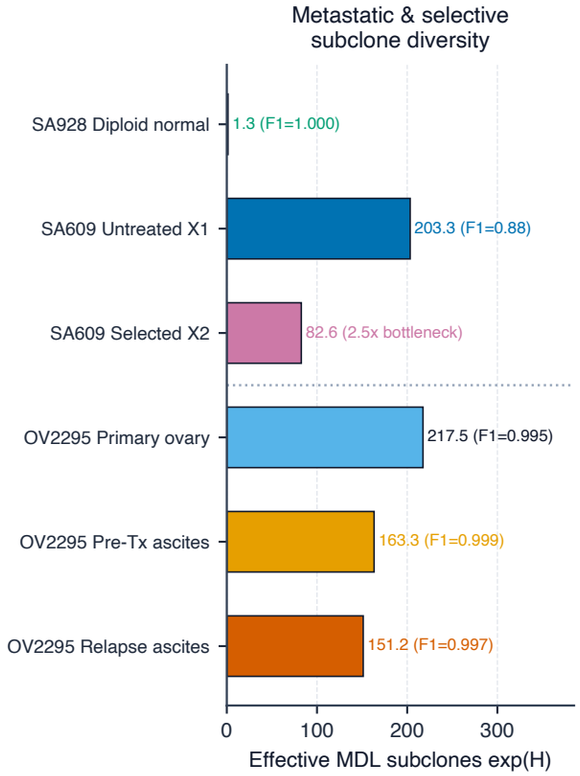
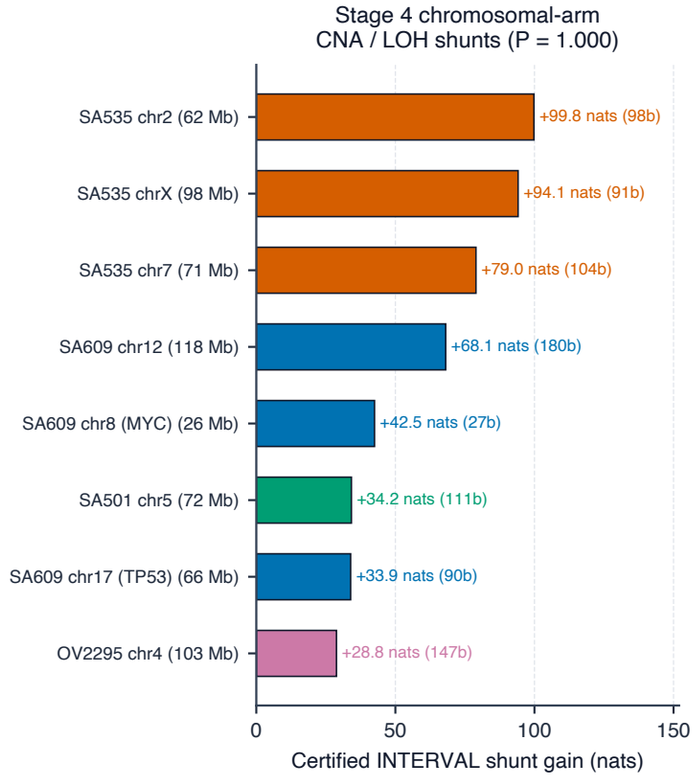
A


scWGS 500-kb CNA compression
(4,950 cells x 4,375 bins, K=12)
Ordered Wagner + RLE (--ordered)
Unordered Fitch (--discrete)
Lossless MDL compression ratio (raw / net)
33.8x
(7.6s)
30.0x
(1.1s)
29.6x
(2.2s)
22.1x
(1.9s)
SA501
(N=2691)
SA535
(N=548)
OV2295
(N=1111)
SA609
(N=600)

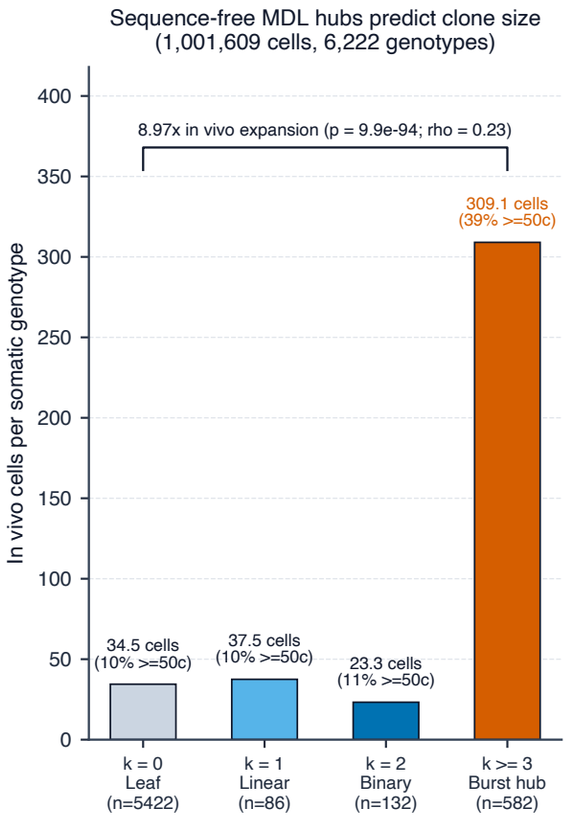

Zero-leakage clade concordance
& false-split contraction (%)
MDL polytomy tree
Forced binary tree
Random permutation
Held-out compartment clade purity
-35% splits
-46% splits
-41% splits
-52% splits
SA501
(N=2691)
SA535
(N=548)
OV2295
(N=1111)
SA609
(N=600)

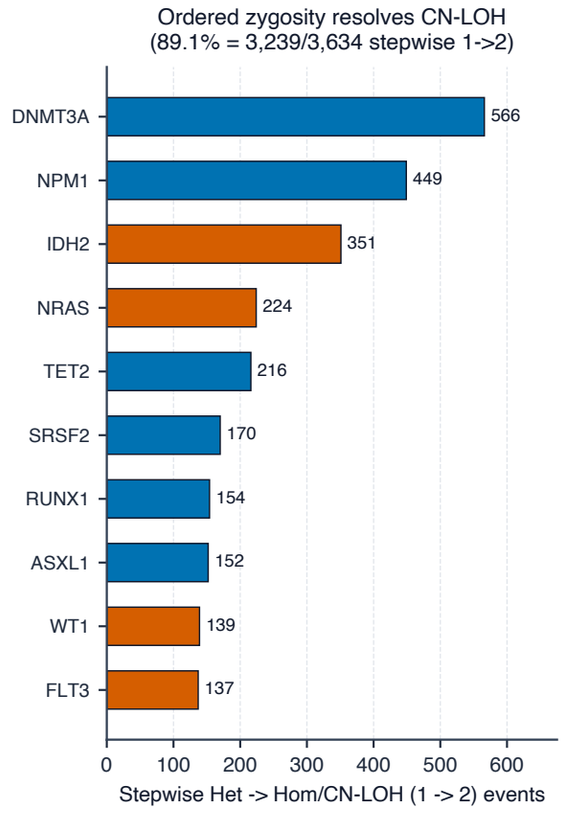
B


Metastatic & selective
subclone diversity
SA928 Diploid normal
1.3 (F1=1.000)
SA609 Untreated X1
203.3 (F1=0.88)
SA609 Selected X2
82.6 (2.5x bottleneck)
OV2295 Primary ovary
217.5 (F1=0.995)
OV2295 Pre-Tx ascites
163.3 (F1=0.999)
OV2295 Relapse ascites
151.2 (F1=0.997)
Effective MDL subclones exp(H)

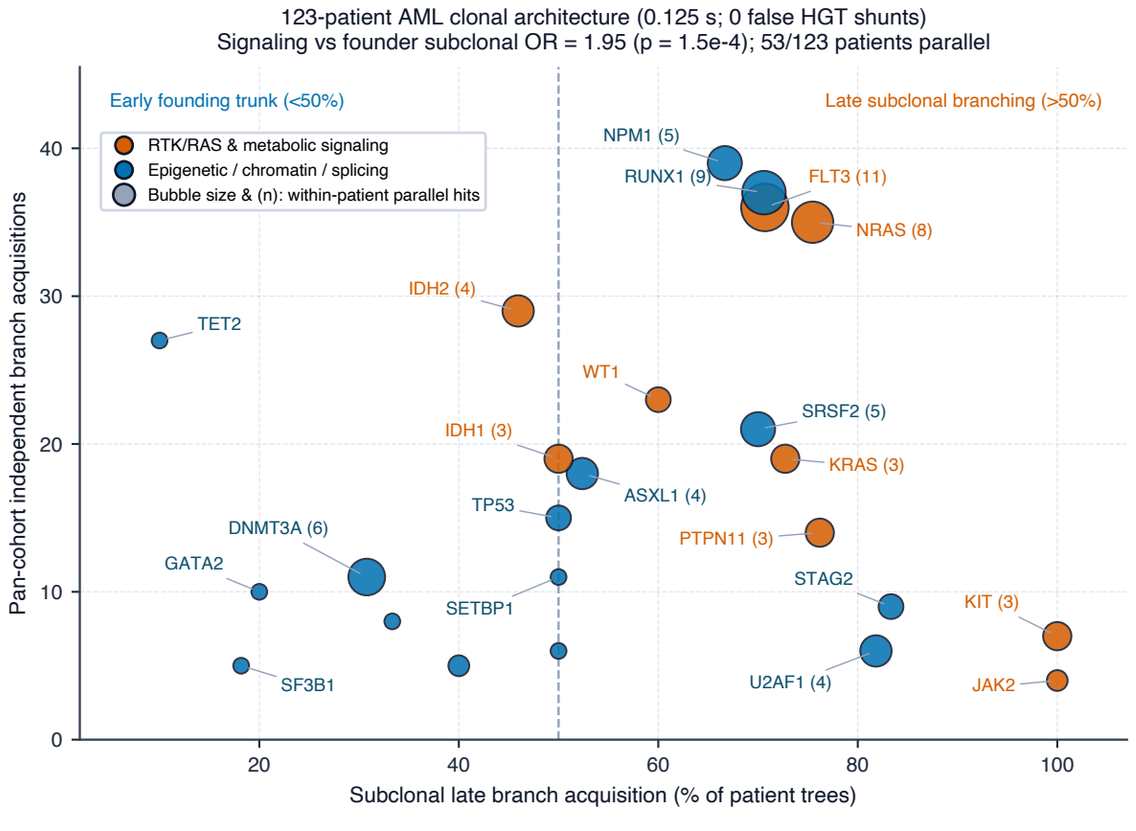

Stage 4 chromosomal-arm
CNA / LOH shunts (P = 1.000)
SA535 chr2 (62 Mb)
+99.8 nats (98b)
SA535 chrX (98 Mb)
+94.1 nats (91b)
SA535 chr7 (71 Mb)
+79.0 nats (104b)
SA609 chr12 (118 Mb)
+68.1 nats (180b)
SA609 chr8 (MYC) (26 Mb)
+42.5 nats (27b)
SA501 chr5 (72 Mb)
+34.2 nats (111b)
SA609 chr17 (TP53) (66 Mb)
+33.9 nats (90b)
OV2295 chr4 (103 Mb)
+28.8 nats (147b)
Certified INTERVAL shunt gain (nats)

C


Sequence-free MDL hubs predict clone size
(1,001,609 cells, 6,222 genotypes)
8.97x in vivo expansion (p = 9.9e-94; rho = 0.23)
In vivo cells per somatic genotype
34.5 cells
(10% >=50c)
37.5 cells
(10% >=50c)
23.3 cells
(11% >=50c)
309.1 cells
(39% >=50c)
k = 0
Leaf
(n=5422)
k = 1
Linear
(n=86)
k = 2
Binary
(n=132)
k >= 3
Burst hub
(n=582)


Ordered zygosity resolves CN-LOH
(89.1% = 3,239/3,634 stepwise 1->2)
DNMT3A
566
NPM1
449
IDH2
351
NRAS
224
TET2
216
SRSF2
170
RUNX1
154
ASXL1
152
WT1
139
FLT3
137
Stepwise Het -> Hom/CN-LOH (1 -> 2) events

D


123-patient AML clonal architecture (0.125 s; 0 false HGT shunts)
Signaling vs founder subclonal OR = 1.95 (p = 1.5e-4); 53/123 patients parallel
Early founding trunk (<50%)
Late subclonal branching (>50%)
RTK/RAS & metabolic signaling
Epigenetic / chromatin / splicing
Bubble size & (n): within-patient parallel hits
Pan-cohort independent branch acquisitions
NPM1 (5)
RUNX1 (9)
FLT3 (11)
NRAS (8)
IDH2 (4)
TET2
WT1
SRSF2 (5)
IDH1 (3)
KRAS (3)
ASXL1 (4)
TP53
DNMT3A (6)
PTPN11 (3)
GATA2
STAG2
SETBP1
KIT (3)
U2AF1 (4)
JAK2
SF3B1
Subclonal late branch acquisition (% of patient trees)